Kyungmin Yu

Los Angeles, CA | +1 213-865-9872 | ykmyou@gmail.com
linkedin.com/in/kyungmin-yu-62b360154 | github.com/KyungminYu | kyungminyu.github.io/Portfolio

Education

University of Southern California — Los Angeles, CA
M.S. in Computer Science — *Aug. 2026 – Expected May 2028*

Sangmyung University — Seoul, South Korea
B.S. in Computer Science; GPA: 4.4/4.5; Rank: 1st in Department — *Mar. 2012 – Feb. 2018*

Experience

Samsung Electronics, MX Division — Suwon, South Korea
Software Engineer (currently on academic leave) — *Mar. 2018 – Present*

Application Development Team – Samsung Notes — *Jun. 2019 – Present*

Systems, Reliability & Performance

- Kept Samsung Notes cold-start latency within 10% of baseline across major releases despite OS changes and continuous feature growth by controlling app size growth, optimizing startup code paths, and rescheduling initialization tasks.
- Improved scrolling responsiveness by optimizing thumbnail loading, resolving memory leaks, and refining the view layer, meeting internal UX validation criteria for release.
- Implemented cross-platform collaborative editing in C++ and JNI, enabling synchronized document editing across Android and UWP devices.
- Integrated a Kotlin/Java multi-connectivity framework for nearby-device task continuity, defining handoff scenarios and preserving compatibility across application updates and data-model migrations.

Applied AI & Production Integration

- Built a Java/Kotlin LLM-to-XML pipeline producing editable Samsung Notes objects across 3 template families and 18 layouts; evaluated prompts in 27 languages using golden sets, manual validation, and a three-criterion rubric, reaching 2.7/3.0 production readiness.
- Built a Java/Kotlin retrieval-augmented document discovery workflow that prepared application context, ranked results for natural-language queries, and captured trial-UX feedback for model improvement.
- Integrated multimodal generation APIs in Java and Kotlin, optimizing preprocessing and postprocessing to deliver results within 15 seconds without sacrificing quality.
- Built a bilingual DistilBERT classifier in Java and Kotlin using 2.4K labeled samples, achieving over 95% F1 and reducing false positives by routing low-confidence results to an Unknown class.

Product Development Team — *Mar. 2018 – May 2019*

Test Infrastructure & Reliability Automation

- Built Java/Bash automation for device pooling and self-healing across a 100-server, 1.8K-device Google certification test farm, improving reliability and reducing certification cycle time by approximately 5x.

Projects

SYNC – Cal Hacks 13.0 | Devpost | GitHub — *Oct. 2026*
Co-Creator & Full-Stack Engineer | TypeScript, React, Phaser, SpacetimeDB, Anthropic Claude API

- Prototyped a multi-agent workflow dashboard in Figma and connected it to live SpacetimeDB state, comparing baseline and SYNC runs through scorecards, decision logs, and nine workload scenarios.
- Integrated the Claude API for workload-aware staffing by role, caching decisions to reduce redundant requests and adding an Algorithm/LLM toggle.

Patents, Publications & Awards

Patents: Contributed to 5 filings in document generation, multimodal GenAI, handwriting interaction, and image presentation (2023–2025).
Publications: 2 peer-reviewed papers: SCIE journal on deep learning for food classification; conference on depth-map object segmentation (2013–2020).
Recognition: Samsung Outstanding Programmer Award (5x), Samsung AI App Specialist, Dean's Award; ACM ICPC Daejeon Regional team participant.

Technical Skills

Languages: C++, Java, Kotlin, Python, TypeScript, Bash
AI/ML: LLM, RAG, DistilBERT, model evaluation, on-device inference, multimodal GenAI
Frameworks: PyTorch, TensorFlow, ONNX Runtime, React, Node.js
Platforms & Data: Android, UWP, JNI, SpacetimeDB, SQL
Systems & Tools: Linux, Git, Docker, performance optimization, memory debugging, reliability, test automation